Assignment 8

Part 1: Mathematical Derivations

Q1

1. we should notice that the standard deviation for $q(x_t|x_{t-1}) = \sqrt{\beta_t}$. the mean is also $\sqrt{1-\beta_t}x_{t-1}$ and so $x_t = \sqrt{1-\beta_t}x_{t-1} + \sqrt{\beta_t}\epsilon_t$ where $\epsilon_t \sim \mathcal{N}(0, I)$

2. given α_t, we can rewrite as

$$\begin{aligned} x_t &= \sqrt{\alpha_t}x_{t-1} + \sqrt{1-\alpha_t}\epsilon_t \\ x_{t-1} &= \sqrt{\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_{t-1}}\epsilon_{t-1} \end{aligned} \tag{1}$$

   which gives

$$x_t = \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\alpha_t(1-\alpha_{t-1})}\epsilon_{t-1} + \sqrt{1-\alpha_t}\epsilon_t \tag{2}$$

   since variances sum, we get

$$x_t = \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\epsilon'_{t-2} \tag{3}$$

   unfolding all the way to x_0, we get

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, \quad \text{where } \epsilon \sim \mathcal{N}(0, I) \tag{4}$$

   which is the same as

$$\mathbf{q(x_t|x_0)} = \mathcal{N}\left(\mathbf{x_t}; \sqrt{\bar{\alpha}_\mathbf{t}}\mathbf{x_0}, (\mathbf{1} - \bar{\alpha}_\mathbf{t})\mathbf{I}\right)$$

3. the distribution just approaches the standard gaussian distribution $\mathcal{N}(0, I)$. as $\bar{\alpha}_T$ goes to zero the mean goes to zero and the variance goes to I

Q2

1.

$$\log p_\theta(x) = \log\left(\frac{e^{-f_\theta(x)}}{Z_\theta}\right) = -f_\theta(x) - \log Z_\theta \tag{5}$$

$$s_\theta(x) = \nabla_x \log p_\theta(x) = \nabla_x\left(-f_\theta(x) - \log Z_\theta\right) \tag{6}$$

   but note that Z_θ is constant wrt to x so the derivative is zero and thus

$$s_\theta(x) = -\nabla f_\theta(x) \tag{7}$$

   but note that Z_θ is constant wrt to x so the derivative is zero.

2. solving for epsilon from part 1,

$$\epsilon = \frac{x_t - \sqrt{\bar{\alpha}_t}x_0}{\sqrt{1-\bar{\alpha}_t}} \tag{8}$$

   substituting

$$q(x_t \mid x_0) = \frac{1}{(2\pi)^{d/2}|(1-\bar{\alpha}_t)\mathbf{I}|^{1/2}} \exp\left(-\frac{\|x_t - \sqrt{\bar{\alpha}_t}x_0\|^2}{2(1-\bar{\alpha}_t)}\right) \tag{9}$$

   and taking the log gives

$$\log q(x_t \mid x_0) = -\frac{\|x_t - \sqrt{\bar{\alpha}_t}x_0\|^2}{2(1-\bar{\alpha}_t)} + C \tag{10}$$

grad of this gives

$$
\begin{aligned}
\nabla_{x_t} \log q(x_t \mid x_0) &= -\frac{2(x_t - \sqrt{\bar{\alpha}_t} x_0)}{2(1-\bar{\alpha}_t)} \\
&= -\frac{x_t - \sqrt{\bar{\alpha}_t} x_0}{1-\bar{\alpha}_t} \\
&= -\frac{\sqrt{1-\bar{\alpha}_t}\epsilon}{1-\bar{\alpha}_t} \\
&= -\frac{\epsilon}{\sqrt{1-\bar{\alpha}_t}}
\end{aligned} \tag{11}
$$

isolating for ε,

$$\epsilon = -\sqrt{1-\bar{\alpha}_t} \nabla_{x_t} \log q(x_t \mid x_0) \tag{12}$$

3. to know the exact probability would require calculating Z exactly which is hard.

Q3

1. (a) ε is gaussian noise,
   t is uniform in $[t]$.
   (b) the network input is a noisy image x_t and a timestep t.
   (c) the network is trying to learn the gaussian noise vector that was added to x_0 to get x_t. MSE penalizes the difference between the prediction and the true noise.
2. (a) σ_t controls the variance of the component added at each reverse step.
   (b) at $t = 1$, this is the final prediction. this step is deterministic of hte input from the prior step
3.

Reflection + Statistics Section

(a)
(b)
(c)
(d) Contributions:

Total:

Part 1:
Q1:

Part 2:
Q1:

Part 3:
Q1:

(e)